

Steering Vision-Language-Action Models via Denoising-Time Guidance in Trajectory Space

Midterm Report

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1. Introduction and Motivation

Vision-Language-Action (VLA) models have recently emerged as a promising paradigm for end-to-end autonomous driving, integrating visual perception, structured reasoning, and continuous action generation into a unified architecture. Alpamayo R1.5 (AR1.5) is a representative example: it produces a Chain-of-Causation (CoC) reasoning trace that explains the causal factors behind a driving decision, then generates a trajectory via a diffusion-based action expert conditioned on that reasoning. The system is explicitly designed so that reasoning and action are coupled — the RL post-training stage optimizes directly for reasoning-action consistency.

This raises a natural question for inference-time control: given that reasoning and action are trained to be consistent, where should behavioral steering intervene: at the reasoning level, the action level, or both?

A natural first approach is to modify the CoC reasoning tokens and observe whether the trajectory changes accordingly. If reasoning and action are tightly coupled, such interventions should produce clear behavioral effects. If they are not, the action expert may have learned to rely primarily on other signals, specifically the visual context encoded in the KV-cache, and effective control may require intervening directly in the trajectory generation process.

In this report, we describe our progress to date. We have conducted a detailed code-level architectural analysis of AR1.5’s forward pass, performed preliminary logit lens experiments on the VLM backbone, and run initial reasoning-level intervention experiments. Based on these findings, we are developing a trajectory-level steering approach that operates within the action expert’s denoising process. Implementation and evaluation of this method constitute our remaining work.

2. Background and Related Work

2.1 Vision-Language-Action Models

VLA models extend vision-language models to embodied action generation. Early approaches such as RT-2 and OpenVLA treat actions as discrete tokens generated autoregressively by the language model. More recent systems adopt diffusion-based action generation, using a separate action expert to produce continuous trajectories. π_0 introduced the dual-module design that AR1.5 inherits: a pretrained VLM backbone

provides semantic context via a KV-cache, and a flow matching action expert generates trajectories conditioned on that cache. AR1.5 extends this with structured Chain-of-Causation reasoning and RL post-training for reasoning-action consistency.

2.2 Diffusion and Flow Matching for Control

Diffusion models generate samples by iteratively denoising noise via a learned score or velocity field. Classifier guidance [Dhariwal and Nichol, 2021] enables control by augmenting the denoising update with gradients of a conditional model. Flow matching [Lipman et al., 2022] provides a continuous-time generalization that learns a vector field transporting noise to data. Applications to trajectory generation in embodied systems remain underexplored, with FlowDrive representing one of the few examples of in-the-loop guidance for driving trajectory flow matching.

2.3 Mechanistic Interpretability

The logit lens [Nostalgebraist, 2020] interprets transformer FFN layers by projecting their output weight vectors onto the vocabulary embedding matrix, revealing which tokens each neuron is most aligned with. Bear et al. [CoRL 2025] apply this technique to $\pi 0$ -family manipulation robots, finding neurons encoding low-level motion primitive concepts and demonstrating inference-time steering via forward hook activation. Our work applies logit lens to AR1.5’s VLM backbone, asking whether CoC training produced higher-level decision-concept neurons beyond kinematic primitives.

2.4 Faithfulness of Chain-of-Thought Reasoning

A growing body of work questions whether chain-of-thought reasoning traces are causally faithful to model outputs. Lanham et al. [2023] show that in language models, modifying reasoning traces often has limited effect on final answers. Turpin et al. [2023] similarly find that CoT reasoning can be influenced by irrelevant biasing features without affecting outputs. These findings directly motivate our investigation into whether AR1.5’s CoC tokens exhibit similar causal unfaithfulness in the trajectory generation context.

3. Architectural Analysis

3.1 Forward Pass Structure

We analyzed AR1.5’s forward pass through direct inspection of its implementation. Given an input observation, the system proceeds as follows.

The VLM backbone processes multi-camera images and ego motion history, autoregressively generating a CoC reasoning trace followed by discrete trajectory tokens, stopping at a special `<traj_future_start>` token. The resulting `past_key_values` (the KV-cache containing keys and values for every processed token across all backbone layers) is then frozen and passed to the action expert.

The action expert is instantiated from the same transformer architecture as the VLM backbone but with modified hidden dimensions and without an embedding layer. It receives the frozen KV-cache as `past_key_values` and iteratively denoises a trajectory

starting from Gaussian noise. At each denoising step, the noisy trajectory is projected into token embedding space via `action_in_proj`, processed jointly with the frozen cache, and projected back to action space via `action_out_proj` to produce a velocity update. The cache is cropped back to its original length after each step via `prompt_cache.crop(prefill_seq_len)`, ensuring the expert always attends into the same fixed context.

A key structural detail from the code is the attention mask applied during expert denoising: positions after the reasoning trace are masked to `-inf`, meaning the expert can only attend up to and including the CoC tokens. It is structurally constrained to use the frozen reasoning context and cannot route around it.

3.2 Structural Observation

This architecture reveals a partial decoupling between two phases: semantic encoding, performed once by the VLM and frozen into the KV-cache; and action dynamics, governed by the iterative denoising process. The reasoning tokens influence the expert only through their frozen cache representation. They are not recomputed during denoising.

This motivates two empirical questions. First, does the KV-cache encode decision-level semantic information the expert can use, or does it primarily encode visual context? Second, how sensitively do trajectory outputs respond to changes in reasoning token content? These questions drive our experiments.

4. Work Completed

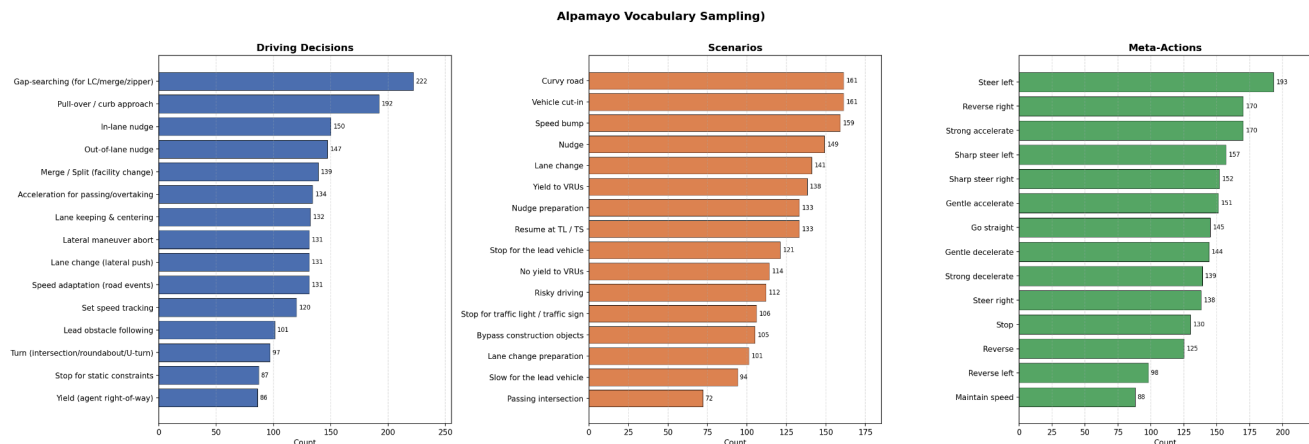
4.1 Logit Lens Analysis of VLM Backbone

We applied the logit lens to AR1.5's VLM backbone FFN layers. For each layer, we extracted the `down_proj` weight matrix and projected each column onto the vocabulary embedding matrix:

```
scores = E · neuron_vector  
top_token = argmax(scores)
```

We focused on neurons whose top vocabulary tokens correspond to the CoC decision vocabulary: yield, stop, nudge, lane change, follow lead vehicle, and close synonyms.

We note this technique is valid only for the VLM backbone, which is vocabulary-trained end to end. It does not apply to the action expert, whose FFN layers are trained to produce denoising vector fields rather than token logits.



Preliminary findings. The resulting distribution across layers is non-uniform, with nudge and lane change tokens appearing as top predictions more frequently than others (Figure above). This is directionally consistent with decision-level neuron specialization induced by CoC training. However, we note an important confound: nudge and lane change scenarios may be more frequent in the training data, producing similar skew without meaningful specialization. We are treating this as an early observation that warrants further investigation rather than a confirmed finding.

4.2 Reasoning-Level Intervention Experiments

To assess the causal role of CoC tokens, we performed initial KV-cache substitution experiments on a small scenario set. We generated caches from two conditions per scenario: the model’s natural reasoning trace, and an alternative trace with the decision component substituted (e.g., “yield” replaced with “accelerate”). We then ran the action expert on both caches and measured trajectory displacement differences.

Preliminary findings. Results were mixed. In some scenarios, substituting the decision token produced only modest trajectory changes, with displacement differences close to the model’s natural sample variance. In others, effects were more pronounced. The relationship between reasoning token content and trajectory output does not appear uniformly strong or weak — it may be scenario-dependent.

We are cautious about drawing conclusions from this limited set. A systematic analysis across a larger and more diverse scenario set is needed, with particular attention to whether effects are larger in challenging long-tail scenarios than nominal ones.

5. Proposed Method: Denoising-Time Guidance

Our preliminary findings suggest that reasoning-level interventions produce inconsistent behavioral effects, which is consistent with prior work on CoT faithfulness [Lanham et al., 2023]. Direct modification of reasoning tokens or the KV-cache also risks producing out-of-distribution inputs the action expert has not seen during training. This motivates a shift toward trajectory-level control.

We propose denoising-time guidance: a steering method that operates entirely within the action expert’s denoising process without modifying the VLM, reasoning trace, or KV-cache. By confining the intervention to the denoising loop, the expert always operates on in-distribution context.

5.1 Classifier Design

We will train a lightweight noise-conditioned trajectory classifier C that takes a noisy trajectory x and noise level t as input and predicts a behavioral class drawn from the CoC longitudinal decision vocabulary. Training will use decision labels from the Physical AI AV Dataset, with noisy samples generated by running the flow matching forward process at various timesteps. We plan to explore three input representations: raw acceleration and curvature profiles following AR1.5’s unicycle dynamics, velocity-derived kinematic features, and learned embeddings from the action expert’s intermediate layers.

5.2 Guidance Formulation

At each denoising step, we will augment the predicted vector field with a gradient term from the classifier:

$$x_{\{t-1\}} = x_t + f_{\theta}(x_t, \text{context}) + \lambda \cdot \nabla_{\{x_t\}} \log p_C(y \mid x_t)$$

Where f_{θ} is the learned vector field, C is the behavior classifier, y is the target behavioral class, and λ is a scalar guidance scale controlling intervention strength.

5.3 Open Questions

Several design choices remain to be resolved through experimentation. We do not yet know whether guidance is most effective applied uniformly across all denoising steps or selectively — for example, stronger in early steps where global trajectory structure is determined and weaker in later refinement steps. We also do not yet know what guidance scale λ balances behavioral steering against trajectory realism, or whether adaptive schedules are needed. These will be addressed empirically.

6. Planned Evaluation

We plan to evaluate denoising-time guidance along four dimensions:

Behavioral alignment. For each target class, the fraction of guided trajectories correctly classified by a held-out classifier. This measures whether guidance successfully shifts the trajectory into the target behavioral region.

Trajectory realism. minADE and minFDE against ground-truth trajectories on the Physical AI AV Dataset test split. Realism should degrade gracefully with increasing λ — strong degradation at low λ would indicate poorly calibrated classifier gradients.

Guidance scale sweep. A sweep of λ plotting behavioral alignment against trajectory realism, characterizing the control-realism tradeoff and the stability of the classifier gradients.

Scenario stratification. Separate evaluation on nominal and challenging scenario subsets. We hypothesize guidance may be more effective in challenging scenarios where the base model’s trajectory is more uncertain.

7. Remaining Work and Timeline

The following work remains:

- Systematic reasoning-level intervention experiments across a larger scenario set with proper quantification
- Classifier training on Physical AI AV Dataset with CoC decision labels
- Integration of guidance into the AR1.5 denoising loop
- Evaluation across the protocol described in Section 6
- Analysis of guidance scale tradeoffs and step-wise guidance schedules

We are currently in the process of setting up the evaluation infrastructure on the Physical AI AV Dataset and expect to begin classifier training within the next two weeks.

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